Sustainability Now!
Sustainability Concepts and Frameworks
Lecture 01

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What is Sustainability?

Sustainability

Sustainability: Concepts and frameworks of stewardship, which are translated to direct human behaviors (decisions and management actions) and applied across multiple, interlinked complex adaptive systems (CAS), to ensure intergenerational equity.

The Goal: Flourishing of the Seven Generations.



Sustainability



Sustainability

Gro Harlem Brundtland, 41

First Woman Prime Minister of Norway, 1981



United Nations

Report of the World Commission on Environment and Development: Our Common Future

Transmitted to the General Assembly as an Annex to document A/42/427 - Development and International Co-operation: Environment

"...development that meets the needs of the present without compromising the ability of future generations to meet their own needs"

Sustainability

Brundtland says:

Wake up! $\rightarrow$ spirit., soc., environ., econ., polit. awareness

Cooperate! $\rightarrow$ empathetic, equity-based cooperation

Break out! $\rightarrow$ imagine new systems and build them



Sustainability

SUSTAINABLE DEVELOPMENT

The central concept of the Brundtland Report, Our Common Future (World Commission on Environment and Development, 1987)

> ***Humanity has the ability to make development sustainable — to ensure that it meets the needs of the present without compromising the ability of future generations to meet their own needs.***
>
> — Our Common Future, Overview, para. 27

THE PRESENT

Meet the needs of today's people — beginning with the basic needs of all

WITHOUT COMPROMISING

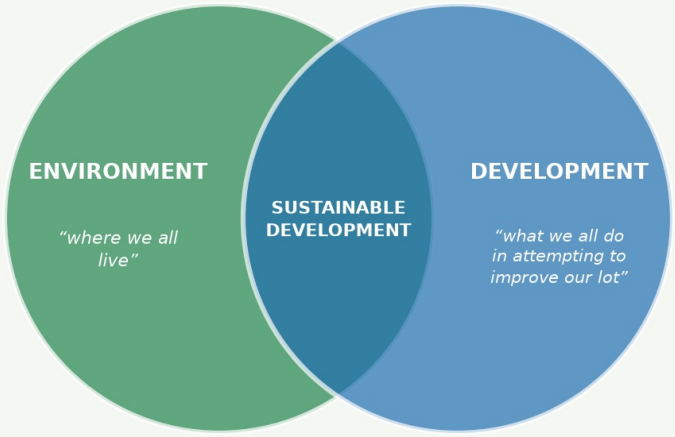
FUTURE GENERATIONS

Preserve their ability to meet their own needs

"We borrow environmental capital from future generations with no intention or prospect of repaying." — para. 25

ENVIRONMENT + DEVELOPMENT — "THE TWO ARE INSEPARABLE"

"Poverty is a major cause and effect of global environmental problems." — para. 8



Environmental, development and energy crises: "These are not separate crises... They are all one." — para. 11

Source of heading quote: Chairman's Foreword

Part 1 of 2 — Continued in Part 2: Three Key Ideas · A Process of Change · Why It Matters

Source: Report of the World Commission on Environment and Development: Our Common Future (United Nations, 1987, A/42/427), Chairman's Foreword & Overview.

Sustainability

SUSTAINABLE DEVELOPMENT

Part 2 of 2 · The Brundtland Report, Our Common Future (World Commission on Environment and Development, 1987)

THREE KEY IDEAS INSIDE THE CONCEPT

NEEDS

Meet the basic needs of all, and extend to all the opportunity to fulfil their aspirations for a better life.

Poverty is an evil in itself, and "a world in which poverty is endemic will always be prone to ecological and other catastrophes."

LIMITS

Not absolute limits, but limits imposed by the present state of technology and social organization, and by the biosphere's ability to absorb the effects of human activities.

Population size and growth must be in harmony with the ecosystem's changing productive potential.

EQUITY

The poor must get their fair share of the resources required to sustain growth.

The affluent must adopt life-styles within the planet's ecological means, for example in their use of energy.

Aided by effective citizen participation and greater democracy in international decision making.

A PROCESS OF CHANGE — NOT A FIXED STATE OF HARMONY

Exploitation of resources → Direction of investments → Orientation of technological development → Institutional change

...all made consistent with FUTURE as well as PRESENT needs. "Painful choices have to be made." — para. 30

WHY IT MATTERS: THE INTERLOCKING CRISES (figures as reported in 1987)

6M ha	11M ha	80×	5 → 8–14 bn	$1 tn
of productive dryland turns into desert every year	of forest destroyed every year	more energy used per person in industrial market economies than in sub-Saharan Africa	world population today vs. projected stabilization next century	global military spending per year

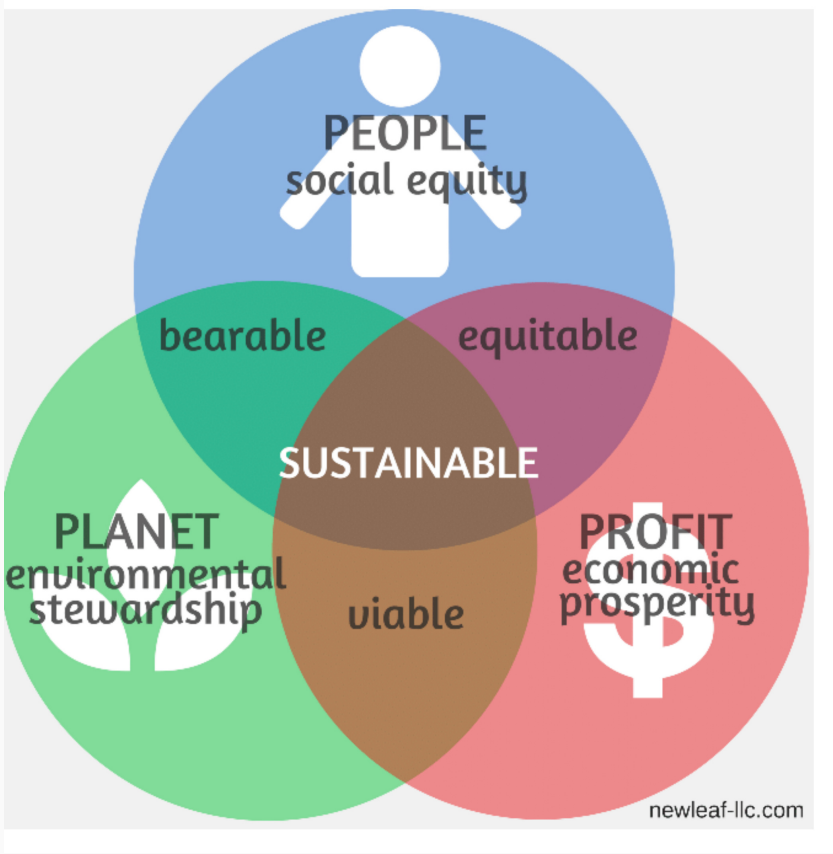
Meanwhile, the gap between rich and poor nations is widening, not shrinking (para. 6).

"In the final analysis, sustainable development must rest on political will."

Who must act: Governments · Private enterprise · NGOs, educators & scientists · People everywhere, especially the young

Source: Report of the World Commission on Environment and Development: Our Common Future (United Nations, 1987, A/42/427), Chairman's Foreword & Overview.

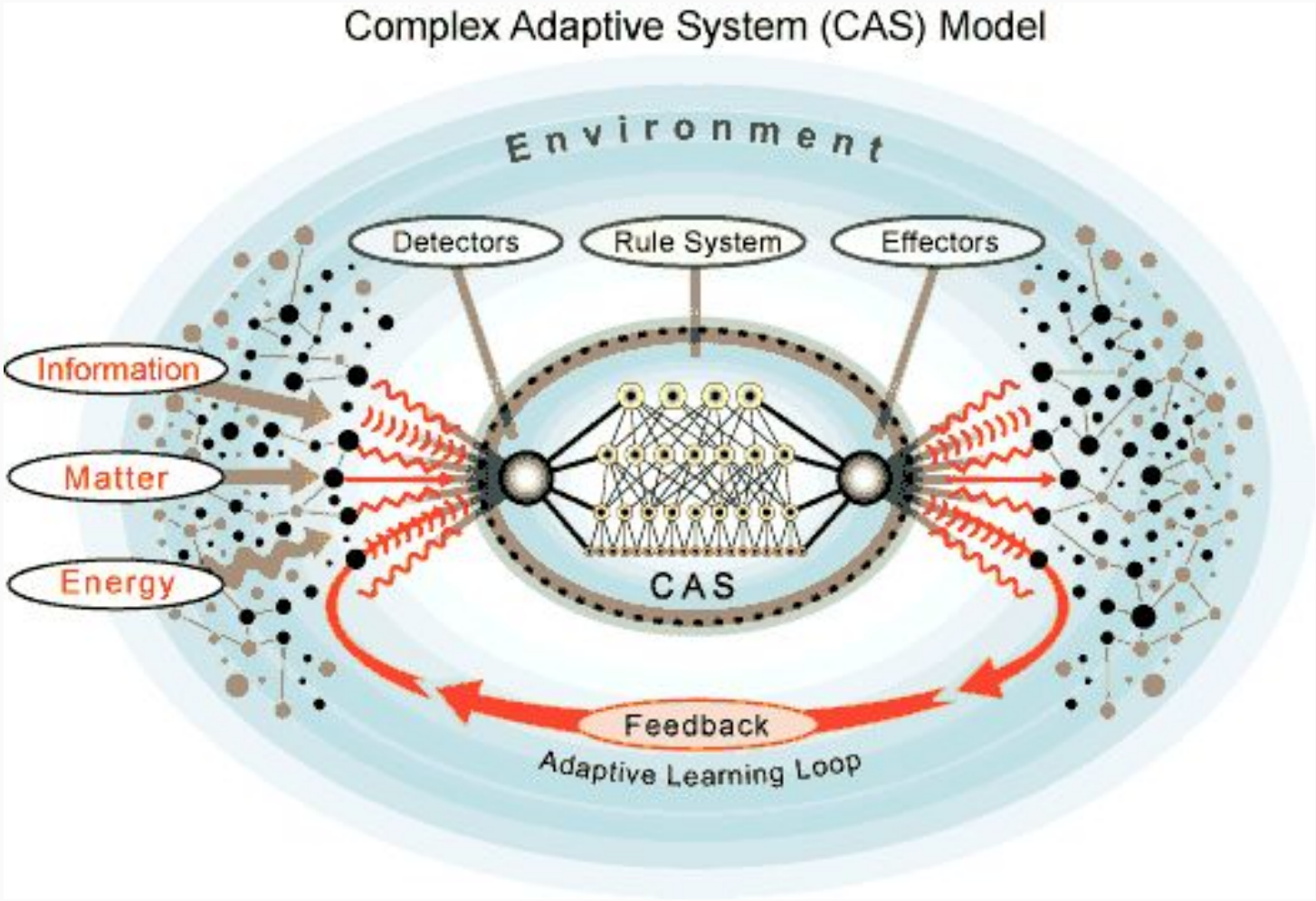
Sustainability



What are the glaring assumptions for this Triple Bottom Line?

What are Complex Adaptive Systems?

Complex Adaptive Systems



Complex Adaptive Systems

Complex means...
Adaptive means ...
System means ...

...Emergence...
...Self-Organization...
...Feedback Loops...
...Resilience...

Complex Adaptive Systems



Complex Adaptive Systems



What is Resilience

Resilience:

The use of dynamic human imagination to adapt, transform, evolve, innovate, self-organize, restructure, and survive when confronted with identity-level disruptive change from turbulence, emergence, collapse, devastation, or crisis.

"The Law of Requisite Variety"

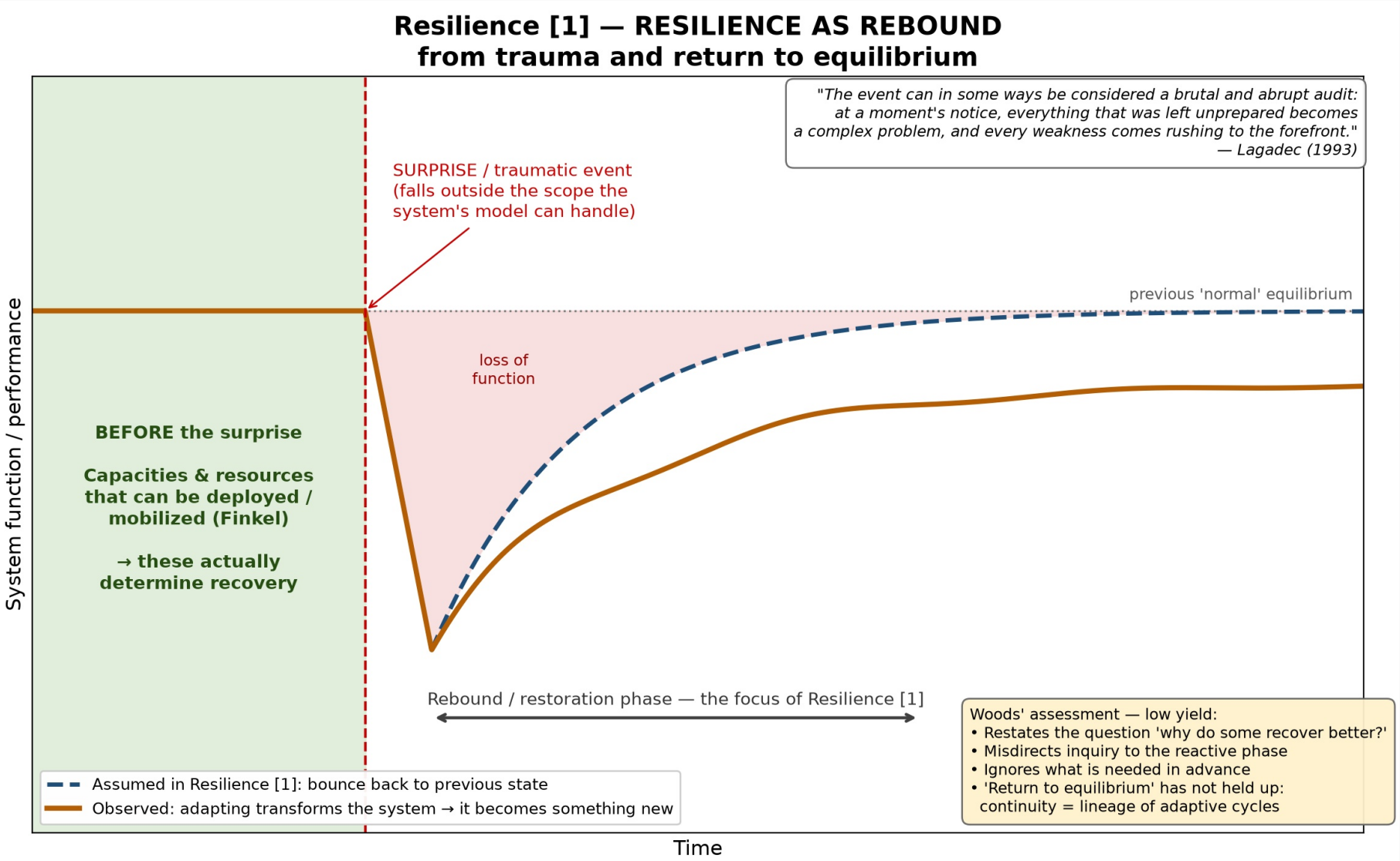
Table 11/3/l

		R		
		α	β	γ
	1	*b*	*a*	*c*
D	2	*a*	*c*	*b*
	3	*c*	*b*	*a*

Table 11/4/l

		R			
		α	β	γ	δ
	1	*b*	*d*	*a*	*a*
	2	*a*	*d*	*a*	*d*
D	3	*d*	*a*	*a*	*a*
	4	*d*	*b*	*a*	*b*
	5	*d*	*a*	*b*	*d*

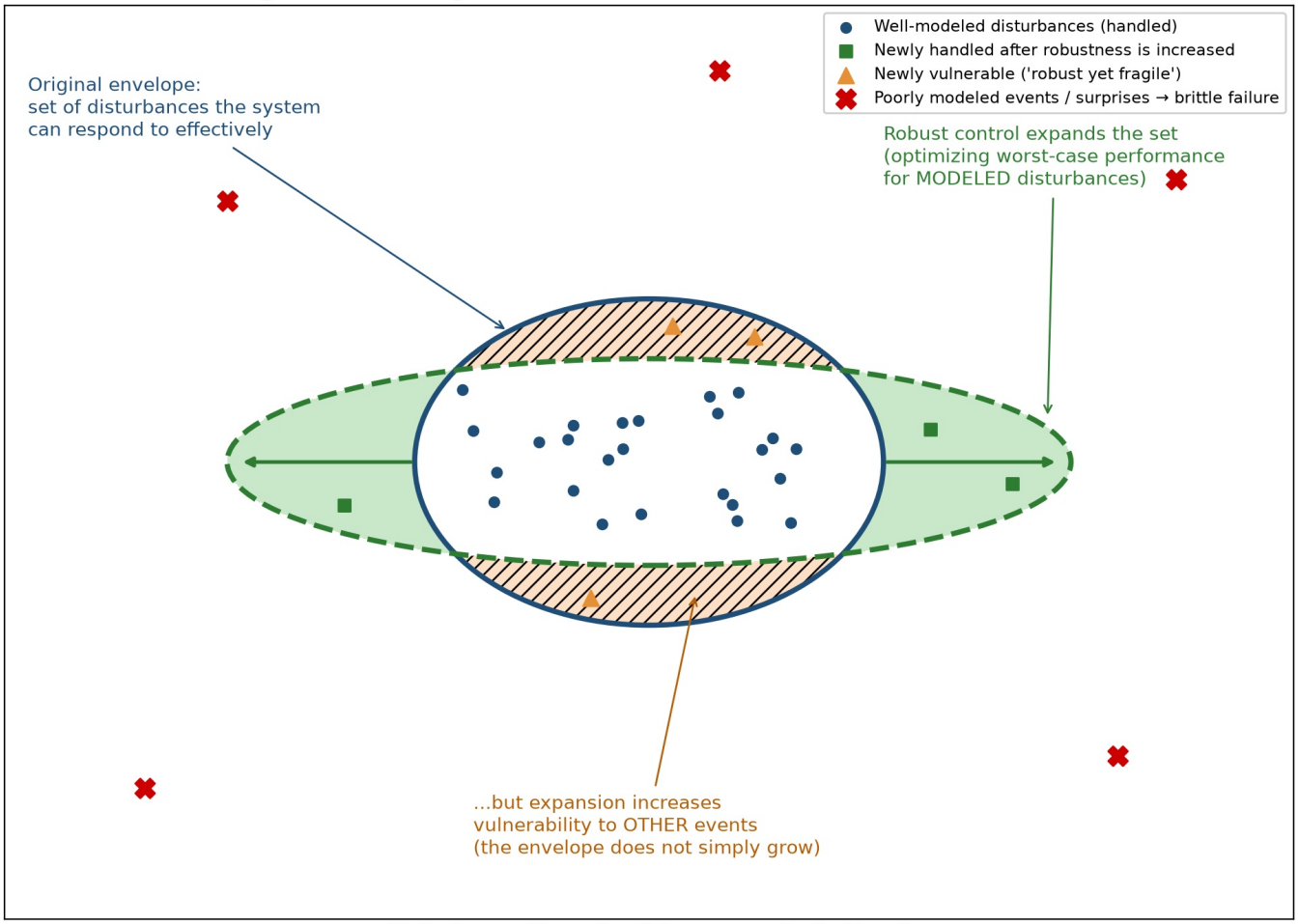
Resilience



s, D.D. (2015). Reliability Engineering & System Safety 141, 5-9.

Resilience

Resilience [2] — RESILIENCE AS ROBUSTNESS
(increased ability to absorb perturbations)




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Robustness always has the form:

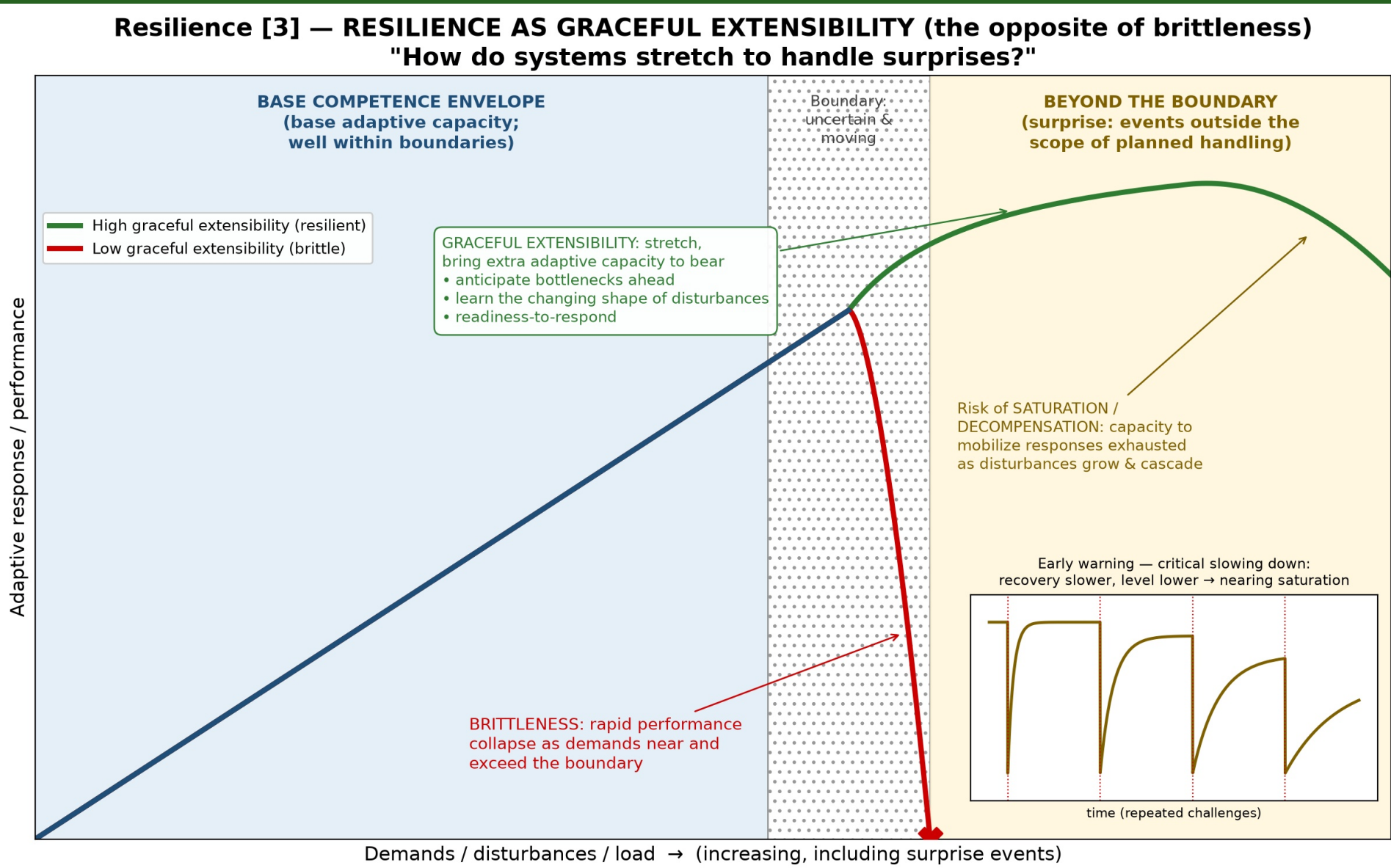
  System X has property Y
  that is robust in sense Z
  to perturbation W
                  – Alderson & Doyle (2010)
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Key points (Woods 2015):

- Robust control works — and ONLY works — when disturbances are well-modeled.
- Fundamental trade-off: becoming more optimal for some disturbances increases brittleness to those outside the set.
- Events outside the set → sudden failure (brittle at the boundaries → Resilience [3]).
- Confounding robustness with resilience begs the question of how systems adapt to poorly modeled, changing events.
- Assessment: low yield; historical value.

e: Woods, D.D. (2015). Reliability Engineering & System Safety 141, 5-9.

Resilience

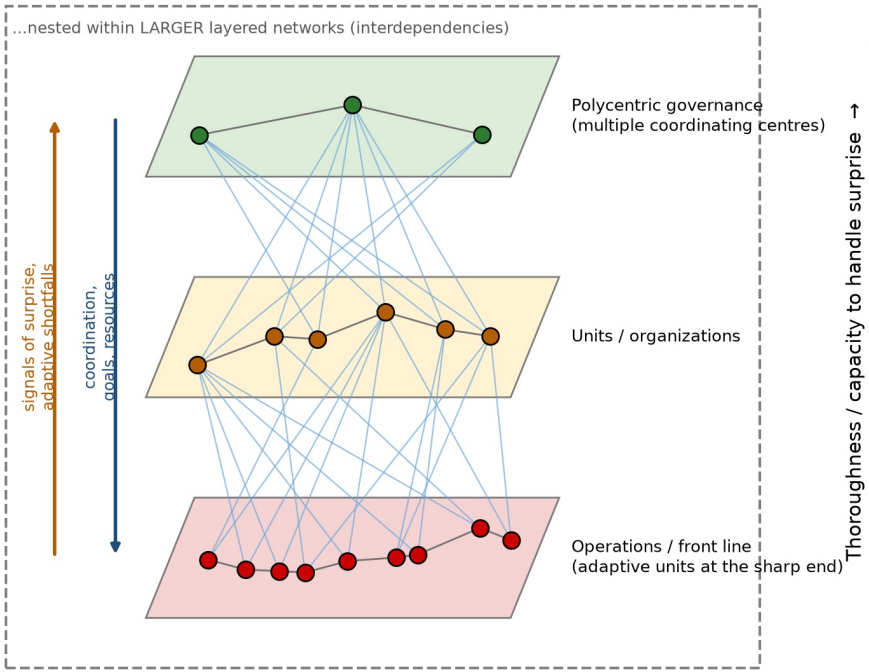


s, D.D. (2015). Reliability Engineering & System Safety 141, 5-9. Plot style after the stress-strain model (Woods & Wreathall, 2008).

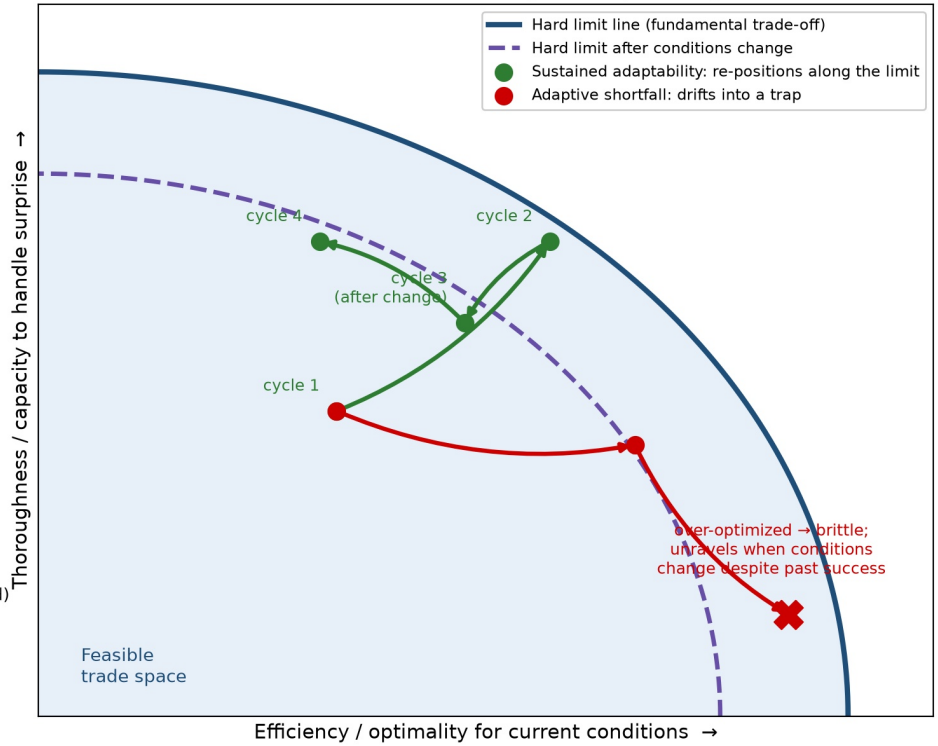
Resilience

Resilience [4] — RESILIENCE AS SUSTAINED ADAPTABILITY
Network architectures (governance policies) that sustain the ability to adapt to future surprises

A. Layered network architecture



B. Trade space over multiple adaptive cycles



Three questions of Resilience [4]:
1. What governance / architectural characteristics distinguish networks that sustain adaptability from those that fail to?
2. What design principles & techniques allow engineering such a network?
3. How can one confidently assess whether a system can sustain adaptability over time?

Exemplars: glycolysis (Chandra et al.), selection for evolvability (Lehman & Stanley), military adaptation to surprise (Finkel), Ostrom's polycentric commons (Bali water temples).

Predictable dynamics of challenge over the life cycle(s):
- Assumptions and boundary conditions will be challenged — surprises re-cur
- Conditions and contexts of use will change — so boundaries will change
- Adaptive shortfalls will occur; people will step in to fill the breach
- The need for, and sources of, graceful extensibility will change — more than once
- The system must readjust itself and its relationships in the layered network

Resilience [4] = managing adaptive capacities relative to multiple fundamental trade-offs.

e: Woods, D.D. (2015). Reliability Engineering & System Safety 141, 5-9.

The Ship of Theseus



Our Home

Mother Earth

